Contents

The Heisenberg Cut as a Physical Threshold: Location, Width, and Consequences of the Classical–Quantum Boundary in Detector Dynamics

Claude Fable 5 (Anthropic) and John M. Bramble, MD

[DRAFT v0.3 — 2026-07-28. Consolidated revision responding to the 2026-07-28 AI review panel (GPT-5.6 Thinking reject 1/5; Gemini major revision 4/5; Grok 4.5 major revision 2/5; internal Fable 5 down-weighted 2/5); per-critique dispositions in the round's authors' response, findings classified against the pre-frozen ledger. Principal changes: the deficit rate renamed κ_{ret} with its ansatz status stated (panel N2, with the sponsor's reversible-return clarification; correction propagated to the companion paper as its v0.5.3); Figure 1 caption and markers corrected — the Hopf and injection-locking boundaries are distinct, as reviewer re-running of the public script established (N1); one operational criterion for the cut (in-principle recoverability) with a consistent engagement/completion/registration time-ordering (N3); $K \sim \Gamma$ demoted to a passive-absorber proxy with the cavity-QED counterexample incorporated; §5 retitled a heuristic correspondence and stripped of ontological assertion; §8.2 demoted to research direction with a preregistrable circuit-QED protocol sketch in its place; capture-reversibility scoped by detector class; the history/diffused-hologram reading added (labeled interpretive, sponsor-originated); authorship-documentation harmonization noted as sponsor action. Reference list verified (19 confirmed / 1 corrected) and §3.3 table sourced row-by-row, as in v0.2. The reviewed v0.2 remains frozen in the round folder. Still pending: final byline; AUTHORSHIP.md/CITATION.cff reconciliation.]

[v0.4 — 2026-09-04. First microscopic test of the recoverability criterion, with predictions fixed before running (`heisenberg_cut_recoverability/`: PREDICTIONS.md, RESULTS.md; ledger open item 7). Four changes, each marked "v0.4" in place. (1) §4.1: recoverability is operationalised as the partial Loschmidt echo — the site's own Hamiltonian reversed, the environment untouched — and distinguished from the coupling-only reversal an experimenter can apply, which fails off resonance even with no environment at all. (2) §3.1: the location claim $\kappa_{\mathrm{ret}}/K = 1$ survives its first exact test, as a rate balance, with the measured midpoint at $\kappa_{\mathrm{ret}}/K = 1.3$–$2.0$; it is restated as a statement about rates. (3) §3.2, §6.2: what the width $w = K/\omega$ is a width of — the layer's extent in share, which in κ_{ret}/K itself is a crossover of relative width order one, as the test measured — and that the on/off of Figure 1(a) is a different observable from the recoverability crossover. (4) §4.1: a dense record channel is a physical requirement for a cut; one record degree of freedom gives coherent exchange, not irreversibility. (5) §8.5, sponsor-originated: temperature as a dial on the record channel — the collapse-in-$\Gamma(T)t$ and fixed-location predictions, with the honest statement that they coincide with Markovian decoherence theory; the falsification list becomes §8.6. (6) §2, lineage paragraph placing the paper against the decoherence programme's own treatment of the cut (Zeh, Zurek, Joos & Zeh, Tegmark, Brune et al., quantum Darwinism, Schlosshauer and Camilleri), with the narrowed statement of what remains distinctively this paper's; and a one-line qualification of §2 item (3)

to the companion's negative result. (7) §4.1: the engaged regime measured in the injected-oscillator model — the stored energy stays at the field-set value through engagement (slope 0.004*) and departs only at the running onset, continuously; "completion" in that model is the running onset, a kink and not a switch. The companion's selection mechanism is now recorded as a negative result (*`NEGATIVE_RESULT.md`*); what this paper imports from it — the slaved-phase statement of §3.1 and the placement of commitment at the vertex in §4.1 — does not depend on the Born-weight derivation that failed.]*

Contributions. Claude Fable 5 (Anthropic) performed the formalization, the manuscript prose, the simulation and figure, and the literature work. John M. Bramble, MD supplied the physical framing and research direction, adjudicated scope and interpretation, and is the accountable human sponsor. Byline order is the sponsor's decision and may be revised before any submission.

Abstract. The Heisenberg cut — the point at which the von Neumann measurement chain is severed and unitary description gives way to definite outcomes — is standardly a bookkeeping convention: movable, unphysical, chosen for convenience, and famously objectionable for exactly that reason. We argue that the cut is a physical threshold with a computable location and a computable width. The argument imports one theorem from a companion paper on outcome selection: below the registration threshold a driven absorber mode has no autonomous phase — it is a globally contracted linear system, its phase slaved to the drive, returning its off-shell excitation to the common field at the rate $\kappa_{\text{ret}} = \Delta E/\hbar$ (a physically motivated ansatz whose microscopic derivation remains open) — while above threshold the mode owns a phase that can lock. The cut sits where the phase-restoring rate falls to the coupling, $\kappa_{\text{ret}} \sim K$, and its fractional width is $w = K/\omega$: sharp beyond resolution for atomic systems ($w \sim 10^{-8}$–10^{-6}), percent-level for broadband solid-state absorbers ($w \sim 10^{-2}$). On this account the celebrated movability of the conventional cut is explained rather than denied — within the reversible sector the placement of the system–apparatus split is provably free, and that freedom hardens into physics only at the locking layer. The framework cleanly separates three notions that measurement talk chronically conflates — entangling interaction, decoherence, and the nonlinear lock — and locates the central interpretive dispute in the third. Consequences follow for the anatomy of a measurement event (capture, selection, registration, with reversibility ending precisely at the layer), for the virtual/real distinction (a heuristic correspondence: off-shell excitation as a failed lock whose lifetime is the energy–time uncertainty; on-shell as locked, with critical slowing at the boundary), and for experiment: the cut has an address in every laboratory, the layer width is a dial that existing platforms already span, and the threshold picture makes coupling — not mass, size, or gravitational self-energy — the variable that decides where quantum behavior ends, a discriminator against both spontaneous-collapse and gravitational-reduction programs. All claims are conditional on the companion paper's single-detector premises. This paper analyses a spatially compact detector, in which all candidate registration sites are causally connected on the timescale of the selection game, and for that geometry it consumes none of the companion's nonlocal ontology — a scoping made explicit in §1 after the companion's v0.7 corrected its own claim that the nonlocal premises are confined to the multi-quantum sector.

---

1. Introduction

Quantum mechanics describes a measurement twice. Inside the formalism, apparatus and system entangle unitarily and nothing ever happens; outside it, an experimenter reads one number off one dial. Somewhere between the two descriptions a line is drawn — system on one side, classical apparatus on the other — and the theory's predictions are indifferent to where. Heisenberg noted the indifference; von Neumann proved a version of it, showing the chain system → detector → amplifier → observer can be cut at any link with identical statistics [von Neumann 1932]; Bohr elevated the line to a principle while declining to locate it. Bell, sixty years later, made the objection canonical: a "shifty split" is no foundation for a physical theory — nothing in physics should depend on a boundary that is nowhere in particular ["Against 'measurement' ": Bell 1990].

The modern consensus assigns the explanatory work to decoherence: interaction with an environment diagonalizes the reduced density matrix in a preferred pointer basis on absurdly short timescales, and the quantum/classical transition is thereby "explained" [Zeh 1970; Zurek 1981; Zurek 1982; Joos & Zeh 1985;

Zurek 2003; Schlosshauer 2004]. The decoherence literature itself is admirably candid that this answers a different question. Diagonality is not an event; a mixture improper for the universe as a whole selects no member of itself; and the program's own reviews state plainly that decoherence does not solve the problem of outcomes [Schlosshauer 2004; Adler 2003]. What decoherence explains is why *interference* becomes unobservable — why the density matrix commutes with the pointer observable — not why *this* pointer position occurs. The cut therefore survives the decoherence program intact: it has merely been renamed the point at which one stops speaking of improper mixtures and starts speaking of occurred events.

This paper claims the cut is physical, locatable, and has a width. The claim is not free-standing: it imports its machinery from a companion paper [Paper 1], which develops outcome selection as a stochastic energy game among detector absorber sites and proves, from stated premises about detectors, that the game is fair and its statistics Born. One theorem of that paper does the work here. **Theorem 2** (restated in §3): an absorber holding energy $e < E_0$ is off-shell by $\Delta E = E_0 - e$ and constitutes a driven *linear* mode returning its excitation at the rate $\kappa_{\mathrm{ret}} = \Delta E/\hbar$; it is globally contracted onto a fixed point whose phase is slaved to the drive; it possesses no zero mode, no autonomous clock, and no capacity for the phase-coherent feedback that irreversible registration requires. Only within a boundary layer $\Delta E \lesssim \hbar K$ of the shell — fractional width $w = K/\omega$, with K the site–field coupling — does an autonomous phase emerge and locking become possible. Far from the shell ($\Delta E \gg \hbar K$): dynamics, interference, reversibility. Inside the layer and beyond it, at the completed lock: commitment. The cut is the layer. (Directional convention, used throughout: "far from the shell" means large deficit ΔE — deeply sub-threshold, deeply reversible; "at the shell" means $\Delta E \to 0$ — the lock's doorstep.)

Three things must be kept distinct, and the paper's architecture follows the distinction. **Settled, and not ours to claim:** the unitary indifference to the placement of the system–apparatus split (von Neumann's theorem, restated in §6 as the *reason* the conventional cut is movable); decoherence theory in full — einselection, pointer bases, the suppression of interference, with its own conclusion that no outcome is thereby selected; and the demonstration by Allahverdyan, Balian and Nieuwenhuizen that *registration* — the amplification of a selected microscopic asymmetry into a stable macroscopic record — is ordinary statistical mechanics of a metastable apparatus [Allahverdyan et al. 2013]. **Claimed:** that the three-way distinction of §2 (entangling interaction / decoherence / nonlinear lock) is physically real; that the lock supplies exactly the element the settled results lack; that the cut is located at $\kappa_{\mathrm{ret}} \sim K$ with width $w = K/\omega$, computable system by system; and that this yields a detector taxonomy, a reading of the virtual/real distinction, and experimental discriminators. **Open:** whether the sub-threshold slaved-phase statement survives full field quantization (inherited from Paper 1's open problems); the closed dynamical model of the lock itself; and everything involving the entangled sector, which this paper does not touch.

That last point deserves emphasis, because it makes this paper's conditionality weaker than its companion's. Paper 1's premises divide into a single-detector core (wave realism P0; capture energetics P1; amplitude-linear coupling P2; detector initial conditions P3; registration taxonomy P4) and a nonlocal extension (shared registry P5; ordering foliation P6).

The scope of that division was corrected in Paper 1 v0.7 (tri-paper round 2, item L2-2), and earlier drafts of this paper inherited the error. Paper 1 previously described P5–P6 as consumed *only* by its multi-quantum and entangled-pair constructions, and this paper repeated that partition. It is false. Exclusivity in Paper 1 is a constraint on an energy ledger closed over all candidate registration sites, so wherever those sites are causally disconnected on the scale of the selection game the ledger must close across the separation — a single photon at a beam-splitter being the simplest case, since otherwise a balanced splitter leaves half a quantum in each arm, neither reaches the full-quantum threshold, and neither ever registers. P5's scope is broadened in v0.7 accordingly, and the nonlocal extension is consumed by any such geometry, not by the multi-quantum sector alone.

What survives for the present paper is the narrower and still-useful statement: **this paper treats a spatially compact detector**, in which all candidate sites are causally connected on the game's timescale, and for that geometry it consumes the core alone. A reader who rejects the nonlocal ontology outright can still evaluate everything here — but that is a fact about the geometry this paper analyses, not about a general exemption held by the single-detector sector.

The paper proceeds as follows. §2 states the three-way distinction. §3 presents the threshold quantitatively and tabulates the layer width across laboratory systems. §4 maps the anatomy of a measurement event onto the threshold and derives the detector taxonomy. §5 reads the virtual/real distinction as sub-/supra-threshold. §6 explains the movable cut — deriving both its freedom and the boundary of that freedom. §7 states the single-world picture and prices it against the alternatives. §8 assembles the experimental consequences. §9 situates the result and lists open problems.

2. Three things not to conflate

Measurement talk chronically merges three physically distinct processes. The central disputes about the measurement problem, in our reading, are disputes about the third; clarity about the first two is what the settled literature already provides.

(1) Entangling interaction. Couple system to apparatus and let them interact. On the joint Hilbert space this is plain unitary evolution: $i\hbar\,\partial_t\Psi = H_{\mathrm{tot}}\Psi$. Entanglement grows; the map is linear and *reversible*. Nothing here distinguishes a measurement from any other interaction — which is precisely von Neumann's point, and the origin of the chain.

(2) Decoherence. Open the apparatus to a large environment and trace it out. The reduced state obeys a completely positive, irreversible master equation — but one *still linear in the density matrix*. Off-diagonal terms in the pointer basis are suppressed at spectacular rates; the state comes to *look* like a classical mixture. But linearity is decisive: a linear map cannot send a superposition to one of its branches, only to a mixture of them, weighted exactly as before. Irreversibility has been purchased; selection has not. Dissipation is not nonlinearity.

(3) The lock. The genuinely nonlinear step: a symmetry-breaking dynamics that takes the distributed, decohered excitation and commits it to one basin — one site, one port, one pointer position. This is where an event becomes an event. Paper 1's content is that the lock exists as detector physics, that its statistics are provably Born under stated premises, and that the noise driving it is the ordinary interference of the site's amplitude with the uncontrolled-phase fields at the detector surface. [v0.4: the companion's Born-statistics claim did not survive its simulations — Paper 1 v0.9 and `NEGATIVE_RESULT.md`; the outcome weights enter there as a premise, and what this paper uses is the lock's existence as detector physics and its threshold structure, not its statistics.] The present paper's content is what the lock's *threshold structure* implies for the geography of the quantum–classical boundary.

In this language the measurement problem decomposes cleanly. Stage (1) is settled unitary dynamics; stage (2) is settled open-system dynamics; the interpretations disagree about stage (3). Everett denies it exists (all basins are occupied); GRW/CSL postulate it as new fundamental stochasticity with new constants [Ghirardi et al. 1986; Bassi & Ghirardi 2003]; Penrose supplies a threshold for it without a mechanism [Penrose 1996]; decoherence-only accounts stop at (2) and redescribe the question. The claim defended here: (3) is ordinary — if nonlinear — detector dynamics, and its threshold is the cut.

Lineage (v0.4). The cut is the decoherence programme's home ground, and this paper's relation to it should be stated exactly. Zeh, Zurek, and Joos and Zeh established that the placement of the split is dynamical, not arbitrary: the environment monitors the apparatus in a pointer basis it selects, and interference between pointer states becomes unobservable at a rate the coupling sets [Zeh 1970; Zurek 1981, 1982; Joos & Zeh 1985]. Tegmark made the consequence explicit as apparent collapse caused by scattering [Tegmark 1993]; Brune and colleagues watched a cavity-field meter decohere progressively as its states were made more distinguishable, a measured crossover of the kind §6.2 describes [Brune et al. 1996]; and quantum Darwinism turned the environment's record into the criterion of objectivity, redundancy of records across environment fragments [Ollivier et al. 2004; Zurek 2009]. Schlosshauer's review states the programme's own limit — decoherence explains the *appearance* of a cut and not the occurrence of one outcome [Schlosshauer 2004] — and Schlosshauer and Camilleri, asking whether decoherence challenges Bohr's doctrine that a classical side must exist, conclude that it justifies where the cut may be placed without making the cut a physical discontinuity [Schlosshauer & Camilleri 2008; Camilleri & Schlosshauer 2015]: the "for all practical purposes" that Bell would not accept [Bell 1990]. Where this paper stands relative to that lineage, after the tests of v0.4: its location claim in κ_{ret}/K is decoherence-compatible, and the recoverability crossover it predicts (§3.1,

§8.5) is what a Markovian environment predicts; the single-outcome claim it inherited from the companion is now recorded there as a negative result (`NEGATIVE_RESULT.md`). What remains distinctively this paper's is narrower than the reviewed draft claimed, and is stated as such: that recoverability ends *in principle* and not for all practical purposes, by the random-reference argument of §4.1, and that the crossing has a coupling-set location with a computable share width. A reader who takes decoherence to have already said everything true here will find the difference in §4.1's trichotomy and §8.6's falsifiers, and nowhere else.

The three-way distinction also disciplines vocabulary that this paper will need. *Reversible* applies to stages (1) and, in the echo/rephasing sense, to the capture stage of §4 — evolution whose effects can be undone by feasible operations. *Irreversible-but-linear* is stage (2): information delocalized beyond practical recovery, statistics unchanged. *Committed* is stage (3) alone. The cut, properly speaking, is the boundary of commitment — not of reversibility (decoherence already ends practical reversibility) and not of interaction (everything interacts). Locating the cut therefore means locating the onset of the nonlinearity.

3. The threshold, quantitatively

3.1 The slaved phase and the layer

We restate the companion result in the form used throughout ([Paper 1] Theorem 2 and Appendix B; proof sketch below, so that no foundational claim rests on an unavailable text). An absorber site short of the full quantum is off-shell: it holds $e < E_0$, with deficit $\Delta E = E_0 - e$, and P4(i) — no stationary state below the transition — makes that deficit a *return rate*, $\kappa_{\rm ret} = \Delta E/\hbar$: the rate at which the virtual (unbankable) excitation reversibly returns to the common field, the beat rate of a failed lock. Two things about this rate, stated before it is used. First, it is not bath dissipation — the returned energy re-enters the exchange pool the selection game redistributes; the symbols γ, Γ are reserved in this paper for genuine dissipative rates. Second, its form is a *physically motivated ansatz*: that the return rate equals $\Delta E/\hbar$, and that reversible give-back may be modeled as a contraction eigenvalue in the site's reduced description, await the microscopic derivation logged in §9 and in the companion's open problems ([Paper 1] §9.4(viii)). Everything below is conditional on the ansatz surviving that calculation — throughout this paper, read "if the mean-field absorber model survives quantization" in front of every quantitative claim. In the rotating frame, with amplitude $a = re^{i\varphi}$, detuning Δ (the drive–transition frequency mismatch, an independent parameter), and drive $fe^{i\varphi_d}$:

$$\dot r = -\kappa_{\rm ret}\, r + f\cos(\varphi_d - \varphi), \qquad \dot\varphi = \Delta + \frac{f}{r}\sin(\varphi_d - \varphi).$$

In Cartesian coordinates this system is *linear*, with eigenvalues $\lambda = -\kappa_{\rm ret} \pm i\Delta$: a unique, globally attracting fixed point, phase offset $\varphi^* = \varphi_d - \arctan(\Delta/\kappa_{\rm ret})$ determined by $(\kappa_{\rm ret}, \Delta)$ alone — identical for identical sites, independent of stored energy. The phase is an enslaved coordinate. No trajectory winds; perturbations decay at rate $\kappa_{\rm ret}$; there are no running solutions and hence no entrainment pull. The site has no clock of its own. (Proof sketch of the companion theorem: linearity in Cartesian coordinates makes the fixed point's global attraction an eigenvalue statement; the absence of running solutions excludes the Adler entrainment average, which is derived from winding trajectories of an *autonomous* limit-cycle oscillator; and P4(i)'s level discreteness prevents banking partial winnings, enforcing threshold sharpness. Full statement and supporting results: [Paper 1] §5.2, Appendix B.)

Dictionary (maintained throughout, and in the companion from its v0.5.3). $\kappa_{\rm ret} = \Delta E/\hbar$: deficit-induced reversible return rate — state-dependent, vanishing at the shell. Δ: drive–transition detuning. K: coherent site–field coupling — the quantity that sets the layer width $w = K/\omega$. Γ: a *measured* coherence-decay rate (linewidth, $1/T_2$, dephasing), a genuinely dissipative quantity used in §3.3 as a proxy for K under stated conditions. $\Delta\omega$: the detuning variable of §5's Adler analysis. Five symbols, five different quantities; the reviewed draft's conflations of them drew justified fire, and the distinctions are load-bearing.

We flag at once the sharpest open formulation, owed to the review panel: the sub-threshold equation above contains no K — the comparison $\kappa_{\rm ret} \lesssim K$ sets a scale, but no single continuous dynamical model yet spans both sides of the boundary and exhibits the bifurcation at $\kappa_{\rm ret}/K = 1$ with coefficients derived from absorber microphysics. Constructing that model *is* the open problem (§9), and the classical oscillator of Figure 1 is an analogy for what its solution should look like, not a substitute for it.

First exact test of the location (v0.4). The smallest model that spans both sides with one continuous dynamics — one photon mode, one absorber whose excitation is detuned by Δ from the shell, so that its off-shell excitation returns at $\kappa_{\mathrm{ret}} = \Delta$, and a dense record channel of 256 modes leaking at a golden-rule rate Γ (`heisenberg_cut_recoverability/`, predictions fixed before running) — shows a crossover, not a bifurcation. Recoverability leaks at Γ times the fraction of the time the excitation spends on the site; that fraction is $2K^2/(\Delta^2 + 4K^2)$, with half-point $\Delta = 2K$, the resonant Rabi frequency, i.e. the rate at which the coupling moves population; and the measured rate-based midpoint is $\kappa_{\mathrm{ret}}/K = 1.3$–$2.0$ across $\Gamma = 0.05$–1 and capture times 3–$10/K$. The location claim holds, with K read as the population-transfer rate, to within a factor 1.6. Two qualifications the test adds. The location is a statement about *rates*: read from recoverability at a fixed observation time it drifts outward, from 1.7 to 5 over $\Gamma t = 0.15$–10, because $\exp(-2\Gamma_{\mathrm{eff}} t)$ reaches a given value at ever smaller occupation. And the crossover's relative width in κ_{ret}/K is of order one (25–75 % points at $\kappa_{\mathrm{ret}}/K \approx 1.2$–$3$), which is the layer width $w = K/\omega$ of §3.2 expressed in the deficit-rate variable, not a separate quantity. The linear model has no autonomous phase on either side, so it tests the location without the bifurcation; whether a self-sustaining absorber sharpens the recoverability crossover is the stage-2 question (§9).

Contrast the autonomous oscillator presupposed by the Adler equation $\dot{\varphi} = \Delta - K \sin\varphi$ [Adler 1946]: amplitude held on a limit cycle by the oscillator's own gain, phase a neutrally stable zero mode, free to run — and, when running, exerting the nonzero average pull that is the engine of injection locking, laser mode competition, and every rich-get-richer instability in synchronization physics [Pikovsky et al. 2001]. The qualitative difference between the two regimes is the theorem: *below threshold there is no zero mode to entrain.* A free clock — the prerequisite for systematic, phase-coherent energy extraction, and hence for winner-take-all feedback and for irreversible commitment — emerges only where the restoring rate collapses, $\kappa_{\mathrm{ret}} \lesssim K$, i.e. within a boundary layer

$$\Delta E \lesssim \hbar K \qquad \Longleftrightarrow \qquad w \equiv \frac{\Delta E_{\mathrm{layer}}}{E_0} = \frac{K}{\omega}$$

of the shell. Figure 1(a) displays the transition as a single observable: an injected oscillator's mean phase-winding rate, swept through threshold. Below, exactly zero — the drive owns the phase. Above, the free clock switches on and runs at the autonomous Adler rate. The crossover is the layer.

3.2 The width, two ways

The primary result is $w = K/\omega$, from marginality: the phase becomes exploitable when its restoring rate falls below the coupling, $\kappa_{\mathrm{ret}} \lesssim K$. The companion's second route — the width–commitment identity, $K \sim \Gamma$ by a fluctuation–dissipation argument, giving $w = \Gamma/\omega$ — is, as the review panel correctly insisted, an *identification, not an identity*: coherent coupling and coherence decay are model-dependently related, with prefactors and spectral-density structure the argument does not supply, and they can be *independently* engineered — cavity QED tunes K by geometry and detuning while holding damping fixed, which is a standing counterexample to any universal $K \sim \Gamma$. The scope of the proxy is therefore stated explicitly: **for passive absorbers** — sites whose coupling to the common field is the same interaction that sets their radiative/dephasing linewidth — $K \sim \Gamma$ is a defensible order-of-magnitude default, and it is what makes the layer width *empirically tabulable* from measured linewidths and frequencies. Where K is engineered independently, the primary formula $w = K/\omega$ applies with the engineered K, and the proxy is dropped. Deriving the precise prefactor for one concrete absorber is a named open task (§9).

What the width is a width of (v0.4). $w = K/\omega$ is the layer's extent in *share*: the deficits $\Delta E \lesssim \hbar K$, i.e. $1 - s \lesssim K/\omega$. In the dimensionless variable κ_{ret}/K the same layer is a crossover of relative width of order one, and the first exact test (§3.1, v0.4) finds exactly that: recoverability changes over $\kappa_{\mathrm{ret}}/K \approx 1.2$–$3$, smoothly, with no sharper structure, and the carrier frequency enters only through the conversion to share. The sharp on/off of Figure 1(a) is a different observable — the winding rate of a self-sustaining oscillator, which switches at a bifurcation — and must not be read as the width of the recoverability crossover, which in a linear absorber with a record channel is smooth. Whether a self-sustaining absorber makes the recoverability crossover itself sharp is open, and is the next test.

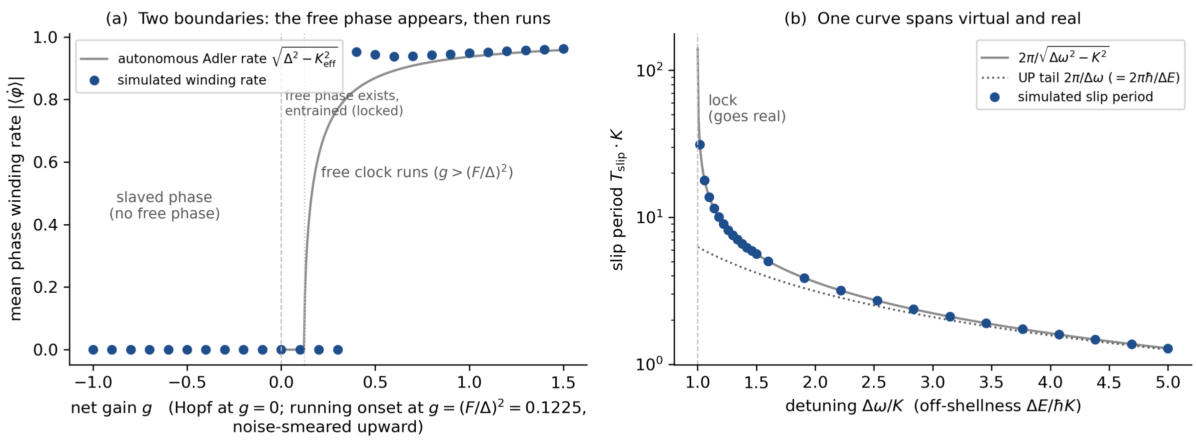

Figure 1: Schematic oscillator analogy — not a simulation of a detector. **(a)** Injected Stuart–Landau oscillator, net gain g swept at fixed detuning Δ and injection F. Two distinct boundaries, both marked: at the Hopf threshold $g = 0$ the free phase comes to *exist* (the linear fixed point gives way to a limit cycle); the winding observable turns on only at the *injection-locking boundary* — the weak-injection Adler estimate puts it at $g = (F/\Delta)^2 = 0.1225$; the model's actual loss of lock is a Hopf bifurcation of the forced fixed point at $g = 0.241$, with the winding onset near $g \approx 0.35$, deterministic and not noise-smeared (v0.4 correction, Appendix A) —, where the free clock can no longer be entrained and *runs*, following the autonomous Adler rate $\sqrt{\Delta^2 - K_{\rm eff}^2}$. The reviewed draft's caption conflated these two thresholds; the correction is owed to the review panel's independent re-run of this script. **(b)** One curve spans the §5 correspondence: Adler slip period versus detuning. Far off-resonance the slip period is the energy–time-uncertainty lifetime $2\pi\hbar/\Delta E$ (dotted); at the tongue boundary it diverges — critical slowing — and the mode locks. Simulated points; analytic curves; methods, convergence, and regeneration command in Appendix A.

3.3 The cut has an address: the layer width across laboratory systems

All rates are angular ($\Gamma = 1/\tau$ for a lifetime τ; linewidths quoted in Hz converted by 2π); each row is anchored to the cited source. **The table is order-of-magnitude, under the passive-absorber proxy $K \sim \Gamma$ of §3.2** — its entries are measured coherence scales converted to layer widths under that stated assumption, not model-independent measurements of the cut.

System	Γ (s^{-1})	ω (s^{-1})	$w = \Gamma/\omega$	character of the cut
Alkali D-line, natural width [Steck 2025]	$3–6 \times 10^7$	$2.1–3.2 \times 10^{15}$	$\sim 1.5 \times 10^{-8}$	razor-sharp; Born exact to all practical resolution
Optical clock transition (^{87}Sr, Al^+) [Dolde et al. 2025; Ludlow et al. 2015]	$4 \times 10^{-3}–5 \times 10^{-2}$	$2.7–7.0 \times 10^{15}$	$2–7 \times 10^{-18}$	sharpest cut in any laboratory
Transmon qubit ($1/T_2$, $T_2 \sim 10–100\ \mu$s) [Kjaergaard et al. 2020]	$10^4–10^5$	$2.5–3.8 \times 10^{10}$	$3 \times 10^{-7}–4 \times 10^{-6}$	atomic-grade sharpness in a macroscopic circuit
Quantum dot, 4 K (radiative $\to$ spectral-diffusion-broadened) [Kuhlmann et al. 2015]	$10^9–10^{11}$	$\approx 2.0 \times 10^{15}$	$6 \times 10^{-7}–5 \times 10^{-5}$	intermediate; the natural dial (§8.2)
Si photodiode / SPAD, 300 K (carrier dephasing) [Sabbah & Riffe 2002]	$4 \times 10^{12}–10^{14}$	$2.7–4.7 \times 10^{15}$	$10^{-3}–3 \times 10^{-2}$	percent-level layer; deviations live here ([Paper 1] §8)

Which Γ, stated explicitly. Each row uses the coherence-decay rate appropriate to the locking analysis — natural (radiative) width for isolated atoms, $1/T_2$ for circuits, momentum-relaxation/dephasing for band carriers — never recombination or readout times. Four caveats travel with the table. (i) The alkali row is the *natural* width; Doppler broadening in room-temperature vapor raises the effective Γ by $\sim 10^2$, toward the $w \sim 10^{-6}$ end quoted for atomic lines in [Paper 1]. (ii) The clock row uses the directly measured ^{87}Sr $^3P_0$ natural lifetime ($\tau = 167$ s); practical clock coherence is laser-limited at $w \lesssim 10^{-15}$, still the sharpest cut ever engineered. (iii) The transmon row is typical circa 2020; current best devices exceed $T_2 \sim 300\ \mu$s–1 ms, pushing $w \to 10^{-8}$. (iv) The quantum-dot upper bound is slow spectral diffusion — inhomogeneous in time rather than a true homogeneous rate — and the silicon dephasing time (32 fs at 5×10^{18} cm^{-3}, 300 K) is density-dependent.

Two readings of the table. First, the cut is *not* at a universal scale: it sits at a different absolute energy in every system, exactly as a convention-based cut would — but its *location in the dimensionless variable* $\kappa_{\rm ret}/K$ is universal, and its width is predicted, not chosen. Second, the table dissolves a familiar puzzle: superconducting circuits are "macroscopic" by any particle-count measure, yet they interfere, violate Bell and Leggett–Garg inequalities, and behave as clean two-level systems. On the threshold account this is unmysterious: their w is atomic-grade. Macroscopicity in mass or particle number is simply not the variable; $\kappa_{\rm ret}/K$ is. §8.4 turns this observation into a discriminator.

4. Anatomy of a measurement event

4.1 Three stages, one crossing

The reviewed draft of this section described the lock as both the commitment boundary and as "engaging reversibly," and the panel correctly identified the collision (finding N3). We repair it with one operational criterion, stated first and used everywhere: **the cut is where in-principle recoverability ends** — the first point in the chain at which no operation, however idealized, can restore the pre-event configuration. *Operationalised (v0.4).* Recoverability is measured as the partial Loschmidt echo — the site's own Hamiltonian reversed for the capture time, the environment left untouched — and not as the reversal an experimenter can apply to the site–field coupling alone. The two differ even with no environment: flipping the coupling returns a detuned capture only on resonance, because the detuning keeps running forward and the flipped coupling tilts the rotation the other way instead of undoing it. In the exact model of `heisenberg_cut_recoverability/` the coupling-flip return fell to 0.04 off resonance while the echo returned the quantum to one part in 10^{15} at every detuning. An experiment that locates the cut by a reversal protocol must therefore separate the two quantities; the cut is defined by the first. (The framework paper restates this criterion in information language — energy is conserved across the cut while clock-position information is not, the two bound to time through $\hbar$ and $k_B T$ as exchange rates rather than identified; Paper 3 §4.2, added 2026-09-02.) Against that criterion, one consistent time-ordering:

this paper (§2)	companion ([Paper 1])	dynamics	recoverability
stage (1) entangling interaction	capture	unitary, distributed excitation	recoverable — demonstrated (echoes, quantum memories)
stage (3) begins	selection: lock *engagement*	fast synchronization; exchange continues	recoverable in principle — the engaged lock can still dissolve (v0.4: now a measured statement — the stored energy stays at the field-set value through engagement, slope 0.004 per unit gain; paragraph below)
the cut	selection: lock *completion*	one site assembles the full quantum (absorption at $s = 1$)	**in-principle recoverability ends here**
stage (2) + amplification	registration	record written by reservoir-powered amplification (the first irreversibility precedes it — Paper 1 v0.8 §6.1)	irrecoverable and now also macroscopic

Capture completes on the fastest timescale the interaction permits ($\tau \gtrsim \hbar/\Delta E$, saturated at $\sim 1/\omega$ for a massless quantum). The lock's *engagement* is reversible — that is the content of the companion's temporal ladder, and the word "lock" in the reviewed draft ambiguously named both phases. The lock's *completion* — the whole stake at one site — is the crossing. Registration, the Curie–Weiss physics of Allahverdyan–Balian–Nieuwenhuizen [Allahverdyan et al. 2013], finalizes and amplifies what completion decided; by the companion's Theorem 4 (commit-rate independence) its speed never touches the statistics — a result whose status Paper 1 v0.7 changed from the operative mechanism for continuum absorbers to a robustness rider, the outcome in that paper's own timescale regime being fixed by first passage before any commit-rate law acts. Paper 1 v0.8 (2026-09-02) withdraws that demotion in turn: with commitment placed at first irreversibility — the interband vertex and thermalisation of an exposed site, 10 fs–1 ps in silicon, not the avalanche — the separation is marginal for silicon and inverted for superconducting absorbers, so Theorem 4 protects the statistics through the linearity of the vertex law (Theorem 5), not through the slowness of commitment. The row above should be read accordingly: the cut is crossed at completion and the record is written later; but in a continuum absorber a site above $\theta = E_{\text{gap}}/E_{\text{photon}}$ can cross before completion, which is the threshold-gated channel Paper 1 v0.8 adds to its ledger. Paper 1 v0.9 then adopts the reading under which no such gate exists — the sub-threshold holding is an amplitude, and commitment is available from every site at a hazard linear in its share — so the row stands as written, with the qualification that "completion" under that reading is the whole-quantum vertex itself, at whichever site fires. The commonplace that "collapse happens when the record is amplified" misassigns the cut by one stage: amplification finalizes; completion decides.

A physical requirement the test adds (v0.4). Irreversibility needs a *dense* record channel. In the exact model, one record degree of freedom coupled to the site gives coherent exchange — the return oscillates in the coupling strength and never settles — four modes leak nothing, sixteen recur within the capture time, and only a channel dense on the capture timescale (256 modes over the channel's bandwidth, recurrence time $2\pi N/B \gg t$) behaves as the golden-rule continuum in which what leaks does not return. The 576-state two-absorber model that serves as Paper 1's unitary baseline, with one record qubit per site, therefore cannot exhibit a cut as built. Stated for detectors: a "record" is a cut only where it is a continuum on the capture timescale — which is the random-reference condition P3(a) of the trichotomy below, restated as a mode count.

The engaged regime, measured (v0.4, from stage 2). The distinction between engagement and completion, which the reviewed draft could only argue, now has a number behind it in the paper's own continuous model (Appendix A; `heisenberg_cut_recoverability/STAGE2_RESULTS.md`). Sweeping the injected Stuart–Landau oscillator's gain through its two boundaries and tracking the stored energy $|a|^2$ — the quantity a record channel leaks, since the exact model of §3.1 showed the leak rate to be Γ times the site's occupation — gives the three regimes in one observable. *Slaved* ($g < 0$): the energy is the linear response $F^2/(g^2 + \Delta^2)$. *Engaged* ($0 < g < 0.355$): a free phase exists from the Hopf point $g = 0$ and the forced fixed point loses stability at $g = 0.241$, yet the stored energy stays at the field-set value F^2/Δ^2 throughout, with slope 0.004 per unit gain — the site holds exactly what the field put in it, and nothing it could leak has changed. *Running* (beyond the winding onset at $g = 0.355$): the energy rises with slope 1.1, tracking the oscillator's own limit-cycle energy g — the bookkeeping has changed hands from field to site. The handover is continuous, with a corner and no jump, and a record channel ($\Gamma_{\text{rec}} = 0.1$) shifts the onset to 0.397 without changing that shape. Two consequences for the table. The engaged row's "recoverable in principle" is now a measured statement: by the energy criterion the engaged lock is on the recoverable side, because it has not yet taken anything from the field that the slaved site did not also hold. And "completion", in this model, is the running onset — the point at which the site's energy first departs from the field-set value — which is a kink in a smooth curve rather than a switch; the sharp switch belongs to the phase-winding observable alone. The model is the classical analogy of Appendix A, with gain put in by hand; the quantum self-sustaining absorber of §9 is where the same three regimes would have to be shown with κ_{ret} a return and not a damping.

Capture's reversibility, scoped honestly (panel finding N5): it is *demonstrated* for controlled, discrete-level ensembles — spin echoes, photon echoes, and atomic-frequency-comb memories run capture backwards on the bench, at the single-photon level, with no registration having occurred [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009] — and *in-principle but undemonstrated, and practically foreclosed,* for broadband continuum absorbers, whose reference structure (below) is uncontrolled. The citations establish rephasing and retrieval; the extension of "reversible capture" to arbitrary detector bulks is this framework's inference, not the cited experiments' content.

Recoverability as history: the reference trichotomy. [Labeled interpretation, sponsor-originated; developed jointly with the companion's v0.5.3 §4.] What recoverability physically *is*: a reversal requires the stored phase history to point back along the path — a wavefront can be run backward only by an agent holding its complete phase record, as phase-conjugate holographic readout demonstrates operationally. Whether the record survives is set by the *reference structure* of the medium:

reference structure across sites	record	consequence
common (hologram, homodyne local oscillator, AFC comb)	invertible	reversible; wavefront reconstructable
different but deterministically known (inhomogeneous ensemble)	invertible by rephasing	echo-recoverable — capture without commitment
random and unrecorded (P3(a) detector bulk; each exchange hop)	non-invertible	history diffuses; commitment possible — the cut's side of the line

The middle row is why echoes work; the reversible/irreversible dephasing split of magnetic resonance is this trichotomy under its textbook name. The bottom row is the companion's P3(a) — the same reference incoherence that makes the selection game fair makes the record non-invertible, so fairness and irreversibility share one origin. In this language the cut criterion above reads: *the cut is crossed when the incident wave's history becomes non-invertible* — diluted across unrecorded references faster than any operation could reassemble it — and Zurek's quantum Darwinism describes the complementary currency: records *of the outcome* proliferate redundantly [Zurek 2003] precisely as records *of the incident wave's history* die. The crossover between the two record types is another face of the cut.

4.2 The detector taxonomy as threshold phenomenology

Two classifications of real detectors follow from the threshold structure, and their cross-product is exhaustive.

By spectrum (P4). *Discrete-level absorbers* (atoms, molecules, ions): no stationary state below the transition, so the full quantum must assemble at one site; the layer is razor-thin (Table, rows 1–2); sub-threshold excitation is virtual and unbankable — level discreteness enforces threshold sharpness. *Continuum absorbers* (semiconductors, metals): final states at all energies above the gap; a commitment channel is open to any site whose share exceeds $\theta = E_{\text{gap}}/E_{\text{photon}}$ (Paper 1 v0.8 §6.1); the layer is wide ($w \sim 10^{-2}$) and the percent-level deviation phenomenology of Paper 1 §8 lives in it.

By record structure. *Event detectors* (photomultiplier, SPAD, single atom): one lock, one record — the selection game runs once. *Track-recording media* (cloud chamber, emulsion, silicon tracker): Mott's problem [Mott 1929] — a single spherical wave produces a straight track — is, in threshold language, a *sequence of partial locks*: each grain along the track runs a selection game whose initial stakes are conditioned by the preceding lock (the wave re-formed around the previous registration), so the sequence is a chain of conditional Born games, each crossing its own layer. Mott's unitary demonstration that the amplitude concentrates on collinear configurations supplies the conditional stakes; the locks supply the events. A track is not one measurement but a correlated cascade of them — which is why track detectors, not interferometers, are where the classical world's *trajectories* come from.

The taxonomy earns its keep twice below: it sorts every entry of the deviation phenomenology, and it blocks a familiar false generalization — conclusions drawn from event detectors do not automatically transfer to tracking media, whose cascade structure is the reason "watching a trajectory" feels classical.

Correction (2026-08-31, tri-paper round 2, item L2-1). Earlier drafts added here that §8 inherits Paper 1's rule that class-(ii) continuum detectors *evade* late-game bias entirely via commit-rate independence. Paper 1 v0.7 withdraws that rule together with the fast-commit reading it rested on: with commitment slow on the scale of the selection game — Paper 1 §6.1's ladder puts the game three to five orders below the commit time — a continuum absorber runs to the absorbing boundary exactly as a discrete-level one does, traverses the same boundary layer, and incurs the same $O(w)$ deviation. Commit-rate independence (Theorem 4) survives as a robustness result, not as an exemption. The consequence for §8 is a widening rather than a loss: the predicted deviation applies to both spectral classes, so the commonest detector type is no longer exempt from the phenomenology this paper sorts.

Further correction (2026-09-02, Paper 1 v0.8). The "three to five orders" cited above was Paper 1 §6.1's ns–μs commit time, which is the record-writing cycle rather than commitment as that paper defines it. Recomputed at first irreversibility, the separation is marginal in silicon and inverted in superconducting absorbers. The widening conclusion stands — both spectral classes traverse the layer and incur the $O(w)$ deviation — and a second, threshold-gated channel is added for continuum absorbers (Paper 1 §8.1, Table 1; §8.6(vii)), gated at $\theta = E_{\text{gap}}/E_{\text{photon}}$ and graded in the photon-to-gap ratio. Commit-rate independence (Theorem 4) survives, now as the load-bearing result for class (ii) rather than as an exemption or a rider.

5. Virtual and real: a heuristic correspondence

(Retitled and rescoped in v0.3, per the panel's convergent finding: what follows is a correspondence between the locking threshold and the virtual/real vocabulary — not a field-theoretic derivation, and no later result rests on it.) The same threshold that draws the cut can be read onto the virtual/real line, and the reading is quantitative in one respect. The dictionary: off-shell energy imbalance ΔE corresponds to detuning $\hbar\Delta\omega$; the lock threshold is the mass shell.

A virtual excitation is an unlocked, off-resonance mode. It slips against the drive and dies as a transient; its lifetime is the slip time of a failed lock. For the Adler phase outside the locking range, the slip period is

$$\tau_{\text{slip}} = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \xrightarrow{\Delta\omega \gg K} \frac{2\pi}{\Delta\omega} = \frac{2\pi\hbar}{\Delta E},$$

which *is* the energy–time uncertainty lifetime — not postulated but produced, as the beat period of an excitation that fails to lock (Figure 1(b), dotted asymptote). At the tongue boundary $\Delta\omega \to K^+$ the slip period diverges: critical slowing. The excitation stops slipping, locks, and persists — the correspondence's reading of "going on-shell." One curve spans both regimes (Figure 1(b)): the far tail reproduces the uncertainty-principle lifetime; the divergence marks the lock; the boundary between them is the same $\kappa_{\text{ret}} \sim K$ layer that §3 identified as the cut. The governing clock changes at the crossing: $\hbar/\Delta E$ (off-shell borrowing) below; infinite (stable attractor) or $\hbar/\Gamma$ (golden-rule decay through open channels) above. For a real but unstable particle the product $\Gamma\tau \sim \hbar$ still *resembles* the uncertainty principle, but as a consequence — the Fourier width of an exponentially decaying amplitude — not as the off-shell borrowing that governs the virtual side.

Status of this section, stated plainly — now in the panel's own terms. In quantum field theory, off-shellness is a statement about four-momenta failing a dispersion relation; virtual excitations are internal contributions to perturbative amplitudes, not observable transients carrying borrowed energy on an uncertainty clock; the energy–time relation is not a license to violate conservation for a prescribed time; and the 2π in the slip-time formula is an artifact of the Adler model, not a QFT theorem. No propagator, spectral-function, pole, or LSZ-type argument connects the locking boundary to a mass shell here — a genuine field-theoretic version would have to derive the detector's Green function and show its poles or spectral support reorganizing at the locking transition, which no one has done. The correspondence is therefore a *re-description with one quantitative consequence*, not an independent derivation: standard quantum field theory computes with off-shell propagators and needs no story about what a virtual excitation "is doing," and "virtual-particle lifetime" is itself a heuristic gloss on a propagator integral. What the re-description adds is the middle of Figure 1(b): a definite, measurable crossover profile — the critical-slowing divergence at $\Delta E \to \hbar K$ — where the standard heuristic offers only the two asymptotes. Systems that can be tuned through the layer (§8.2) can in principle trace the middle of the curve; that, and not the asymptotes, is the falsifiable content. We flag the section's altitude accordingly and rest no later result on it.

6. The movable cut, explained

6.1 Why the conventional cut is movable: a theorem, not an embarrassment

Within the reversible sector, the placement of the system–apparatus split is provably free. Von Neumann's chain argument is exactly this: model the detector quantum-mechanically or classically, cut after the system, after the detector, after the amplifier — all choices yield identical statistics, because the intervening dynamics is linear and the eventual outcome weights are basis-quadratic either way [von Neumann 1932]. (Assumptions carried by that argument, stated since we lean on it: the intervening links evolve unitarily; the pointer coupling is of the measurement form that correlates without disturbing the measured basis; and the Born weights are applied, wherever the cut falls, to the same Schmidt coefficients — it is a statistics-equivalence theorem, not an ontology.) The threshold account *keeps this theorem and explains its domain.* Every link of the chain that lies above the layer — every stage whose dynamics is stage-(1) unitary or stage-(2) linear-dissipative — can be placed on either side of the description at zero cost, precisely because linear dynamics commutes with the bookkeeping. The freedom to move the cut is a symptom of linearity, and linearity is exactly what holds *everywhere except the layer.*

The freedom ends where the nonlinearity begins. The lock cannot be placed on the "described unitarily" side of the split, because no unitary (or linear-dissipative) description reproduces commitment: a linear map preserves the superposition's weights and can only deliver mixtures (§2). The layer is therefore the unique link of the chain at which the two descriptions cease to be intertranslatable — the one place the cut *must* go if the description on each side is to be dynamically accurate. The conventional cut's shiftiness and the physical cut's definiteness coexist without tension: the former is the linearity theorem, the latter is the locking threshold, and the confusion between them dissolves once stages (1)–(3) are kept separate.

Bell's complaint is thereby answered in its own terms. The split is shifty *within the sector where nothing depends on it* — that is a theorem, and no foundation should be built on resisting a theorem. It is pinned — located at $\kappa_{\mathrm{ret}} \sim K$, with width $w = K/\omega$ — exactly where something *does* depend on it. Nothing fundamental is nowhere in particular.

6.2 The cut as a crossover

A boundary with a dimensionless location and a finite width is, in the language of many-body physics, a *crossover*, and the naming buys discipline. The location is $\kappa_{\mathrm{ret}}/K = 1$; the width is $w = K/\omega$; (the layer's extent in share — in κ_{ret}/K itself a crossover of relative width order one, measured in v0.4, §3.2); on one side one set of laws dominates (reversible, interference-preserving, split-placement-free), on the other a different term dominates (dissipative, committing, split-pinning). Nothing discontinuous happens to the laws themselves — the locking term is present in the dynamics throughout, and simply has nothing to grip while the phase is slaved. A crossover, not a phase transition; a dominance boundary, not a change of legislation.

Two consequences of the naming. First, *dose–response*: observables that probe the boundary should vary smoothly with the crossover variable κ_{ret}/K, with all deviation phenomenology scaling with the layer width — which is exactly the structure of Paper 1's deviation ledger (deviations $O(w)$: negligible for atomic detectors, percent-level for broadband solids). A conventional cut predicts no such systematics; a crossover mandates them. Second, *completion elsewhere*: in the parent framework the dissipative sector is also where a preferred foliation enters the dynamics, so the same crossover marks where frame-sensitive physics switches from present-but-decoupled to dynamically active. That identification, its symmetry accounting, and its experimental program belong to the framework paper and are deliberately not developed here; we note only that the cut's crossover structure is what makes such a completion well-formed [Paper 3, in preparation].

7. Single world: the occupied basin

The threshold picture yields a one-world reading of the formalism. Above the layer, the wave explores; superposition is physical, interference is real, and *no basin is occupied* — the selection game has not run. At the layer, symmetry breaks: noise — the surface-interference noise identified in Paper 1 — tips the distributed excitation into one basin, and the system *occupies* it. The other basins do not become other worlds; they become unoccupied attractors, exactly as the untaken well of a buckling beam is not a parallel beam. Where Everett reads the persistence of the other branches' amplitudes as their reality, the threshold account reads their amplitudes as *stakes that lost* — physically real before the game (P0 wave realism is not negotiable here), physically redistributed by it (the energy went to the winner; Paper 1 §4).

Pricing against the alternatives, with our own costs on the table:

Program	supplies an event?	mechanism?	new constants?	principal cost
Decoherence-only	no	—	no	redescribes the question; cut survives renamed
Everett/MWI	no *single* outcome (branching; "event" in this paper's sense)	—	no	all branches real; Born weights via decision theory (its adequacy contested both ways)
GRW/CSL	yes	postulated noise	yes (λ, r_C)	new physics engineered for the job; anomalies predicted [Bassi & Ghirardi 2003; Bassi et al. 2013]
Penrose OR	threshold only	none offered	gravitational scale	location without dynamics [Penrose 1996]
ABN (Curie–Weiss)	registration only	statistical mechanics	no	selection assumed at the crucial step [Allahverdyan et al. 2013]
This program	yes	detector locking (Paper 1)	no	conditional on P0–P4 (incl. wave realism); κ_{ret} ansatz and lock dynamics not yet derived; Theorem 2 linear-semiclassical, quantization open

The comparison is deliberately double-edged. Against GRW/CSL: no new constants, no engineered noise — the fairness is proved, not tuned, and the "collapse rate" is the physical commit rate of an actual detector. Against Penrose: a mechanism where he supplies only a scale — though his structural instinct, that the boundary is physical and dynamical rather than conventional, is the same instinct this paper formalizes. Against MWI: one world, at the price of wave realism and the two open problems we carry visibly (the field-quantized sub-threshold statement; the closed lock dynamics). Against decoherence-only: we accept every einselection result wholesale and add the one thing its own reviews concede is missing. And we weight our own column the way the panel insisted it be weighted: "no new constants" is genuine, but wave realism, the κ_{ret} ansatz, the unclosed lock, and (for the entangled sector, elsewhere) a preferred foliation are non-trivial ontological and theoretical costs relative to decoherence-only or Everett — the advantage claimed is not cheapness but *checkability*: this program pays in premises about detectors, which laboratories can interrogate, rather than in metaphysics, which they cannot.

8. Where the cut bites: experimental consequences

8.1 Inherited: the deviation phenomenology scales with the layer

Everything in Paper 1's deviation ledger is, in the present language, layer phenomenology: deviations from exact Born statistics are $O(w)$, hence invisible for atomic-grade detectors and percent-level where $w \sim 10^{-2}$ (broadband solids), concentrated in configurations that couple to the layer (mismatched ports; pre-cohered absorbers). We do not re-derive that ledger; we re-read it as the dose–response curve of the crossover (§6.2):

the same experiments that test Paper 1's selection mechanism test this paper's cut location, because the deviations are the layer made visible.

8.2 The dial: systems engineered to straddle the layer

The table of §3.3 spans sixteen orders of magnitude in w — from an optical-clock transition at 10^{-18} to a room-temperature photodiode at 10^{-2} — and, decisively, some platforms can *tune* it: a quantum dot's linewidth varies orders of magnitude with temperature and phonon coupling; a transmon's dissipation is engineered via its Purcell environment; cavity QED sets K directly by detuning and geometry. The reviewed draft claimed here that boundary phenomenology should vary "with κ_{ret}/K and nothing else" and called it the paper's primary novel experimental claim. The panel's objection is accepted in full: stated at that altitude, the claim is close to dimensional analysis on coherence phenomenology, which standard open-system theory also organizes by coupling and linewidth ratios; real devices carry bath structure, drive, temperature, geometry, and disorder as separate axes; and no proposed-theory curve yet exists to compare against a null model. **The one-parameter collapse is therefore a research direction, not a prediction**, pending the crossover-profile calculation of §9.

What replaces it is a protocol sketch, offered in the preregistrable format the panel specified. *Platform:* a transmon coupled to a lossy readout mode — the one system where the two rates are independently engineered (K by coupler geometry/detuning; dissipation by Purcell filter). *Sweep:* hold the qubit and drive fixed; step the engineered coupling so the ratio crosses the putative layer. *Observables:* (i) the distribution of commitment latencies (time from drive arrival to irreversible readout excursion) and (ii) the recoverable-coherence window (echo-recoverable fraction versus delay), both as functions of the swept ratio. *Null model:* standard circuit-QED master-equation predictions for the same sweep — both observables vary smoothly, with scales set by the engineered rates, and nothing distinguished happens at any particular ratio. *Framework curve:* to be supplied by the §9 calculation; the qualitative expectation is a kink or crossover in both observables co-located at a single ratio, with width tracking w. *Confounds to be budgeted:* thermal occupation of the readout mode, drive-power dependence, non-Markovian filter response. *Status:* the null model is computable today; the framework curve is not yet — which is the honest reason this section is a direction and not a claim. Once that curve exists, the cross-platform version (two systems with matched ratio and wildly different mass, size, and particle number showing matched boundary phenomenology) becomes the discriminating form.

8.3 Macromolecule interferometry, re-read

The interference of large molecules — C_{70}, functionalized oligomers, and by 2019 a 25 kDa, 2000-atom molecule [Arndt et al. 1999; Hornberger et al. 2012; Fein et al. 2019] — is standardly narrated as a march up the mass axis toward an anticipated breakdown. The threshold account predicts the march continues indefinitely *as long as the interfering system stays above its layer*: mass is not the variable. Visibility is lost to decoherence (thermal emission, collisional which-path labeling — stage-(2) physics, fully accounted by the standard theory [Hornberger et al. 2012]) and *committed* only when a detector-grade lock occurs. The discriminating structure is recoverability: stage-(2) visibility loss is in-principle reversible (echo-type recovery; capture-without-registration, §4.1), stage-(3) loss is not. An experiment that could rephase after a putative "collapse" — recovering visibility that CSL or gravitational reduction says is gone for good — separates the programs cleanly: spontaneous-collapse theories predict a mass-scaled, unrecoverable loss channel; the threshold account predicts zero unrecoverable loss until coupling to an above-threshold system, at any mass.

8.4 Leggett–Garg tests as cut-location experiments

Macrorealism, as operationalized by the Leggett–Garg inequality [Leggett & Garg 1985; Emary et al. 2014], asks whether a system has a definite state at all times. The threshold account's answer is sharp: definiteness is *purchased at the layer*, so LGI violations should persist for any system maintained below threshold ($\kappa_{\text{ret}} \ll K$... i.e. kept coherent, weakly monitored) regardless of macroscopicity, and disappear exactly when the monitoring chain includes an above-threshold lock. Superconducting circuits — macroscopic by particle count, atomic-grade by w (Table §3.3) — are the existing confirmation: they violate LGI robustly. The discriminator against mass-scaled programs recurs: CSL predicts LGI violations fail at a mass/size scale independent of coupling; the threshold predicts they fail at a coupling scale independent of mass. The two axes are experimentally orthogonal, and platforms exist along both.

8.5 Temperature as a dial on the record channel: a collapse test (v0.4)

[Sponsor-originated, 2026-09-04. The prediction is derived from the exact model of §3.1 (v0.4) and is stated here before any measurement beyond the published temperature laws cited has been consulted.]

The physics. Thermal motion in the absorber's medium delivers phase kicks to the captured state at random, unrecorded times — the bottom row of §4.1's trichotomy, made by heat. The record of the incident wave's history spreads over e^{S/k_B} configurations of the medium, where S is the entropy produced; S grows at the dephasing rate, so the count grows exponentially in time. One number: a 500 nm photon thermalised at 300 K produces $E/k_BT \approx 96$ units of entropy, spreading its record over $\sim 10^{42}$ microstates within the thermalisation time, and Landauer's bound then prices the reversal — exporting that entropy at temperature T costs at least $T\,\Delta S = E$, the photon's own energy [Landauer 1961]. This is Paper 3 §4.2's triad with numbers on it: k_BT is the exchange rate that converts the lost history into exactly the deposited energy. Two refinements. Time enters as rate × time: echo fidelity decays as $\exp(-t/T_2)$, and the polarisation-echo experiments of Levstein and Pastawski exhibit the strong form, in which beyond a point the decay rate is set by the medium's own many-body dynamics rather than by the imperfection of the reversal, so that a better reversal does not help [Levstein et al. 1998; Pastawski et al. 2000]. And temperature is a proxy: the physical variable is the rate $\Gamma(T)$ of unrecorded kicks, rising with T through phonon occupation in a solid, as $n\sqrt{T}$ through collisions in a gas, and *falling* as $T^{-3/2}$ through Coulomb collisions in a plasma.

What is already known. Every existing temperature study agrees with the picture. Optical coherence in rare-earth-doped crystals — the platform of the echo memories cited in §4.1 — has T_2 near a millisecond at 2 K, collapsing by ~ 10 K under two-phonon Raman (T^7) and Orbach (activated) laws [Equall et al. 1995; Könz et al. 2003]; matter-wave interference of C_{70} fades as the molecules' internal temperature rises and they emit thermal photons that record their path [Hackermüller et al. 2004] — heat writing the record, directly; transmon coherence falls with temperature through activated quasiparticles [Kjaergaard et al. 2020]. None of these is a test of this paper; they fix the qualitative content.

The prediction. The exact model of §3.1 (v0.4) gives recoverability $R = \exp[-2\Gamma\, t\, f(\Delta)]$, with $f(\Delta) = 2K^2/(\Delta^2 + 4K^2)$ the fraction of the capture time the excitation spends on the site, and found R independent of the record channel's bandwidth at fixed Γ to one part in 10^3. Reading the record channel's rate as $\Gamma(T)$, two separable statements follow. (i) **Collapse.** Echo-retrieval efficiency versus storage time, measured at several temperatures, falls onto a single curve when time is scaled by $\Gamma(T)$, provided the channel is memoryless on the capture timescale; a family that does not collapse measures non-Markovian structure in the medium's phonon spectrum (the Markov ratio of the companion's E-16 ledger entry) and is informative rather than fatal only if that structure is independently measured. (ii) **Fixed location.** The retrieval crossover in input detuning sits at $|\Delta| \approx 2K$, the coupling's population-transfer rate, and does not move with temperature while $\Gamma(T) \ll K$; as $\Gamma(T)$ approaches K it may move *inward* by at most a factor 1.5 (the exact model: κ_{ret}/K from 2.0 to 1.3 over $\Gamma/K = 0.05$ to 1) and never outward. Only the *depth* of the crossover is temperature's to set.

Protocol. Platform: an atomic-frequency-comb or two-pulse photon-echo memory in $Pr^{3+}{:}Y_2SiO_5$ or $Eu^{3+}{:}Y_2SiO_5$, where $T_2(T)$ is tabulated and K is set by the comb's optical depth and the input pulse; alternatively the quantum-dot dial of §8.2. Sweep: input detuning across $\pm 6K$ at three or four temperatures spanning a decade in $\Gamma(T)$ (2–10 K in the rare-earth case), and storage times spanning a decade. Observable: retrieval efficiency as a function of (Δ, t, T). Null model: standard Markovian decoherence of a two-level absorber, which predicts the same collapse and the same fixed location — this paper's temperature content coincides with it, and the test cannot discriminate the framework from decoherence theory. Its value is that the framework *owes* a definite temperature prediction and this is it: the cut's location is coupling-set, not temperature-set; cooling widens the reversibility window without moving the crossing (the companion's E-16 result for commitment, restated for recoverability). Falsified by: a retrieval family that neither collapses in $\Gamma(T)\,t$ nor is explained by an independently measured non-Markovian bath spectrum; or a detuning crossover that moves with temperature by more than a factor 1.5, or outward at all, while $\Gamma(T) < K$.

8.6 What would falsify this paper

(i) A system demonstrably held above its layer ($\kappa_{\text{ret}} \ll K$, autonomous phase confirmed by injection-locking phenomenology) that nonetheless shows *unrecoverable* interference loss scaling with mass or size — the CSL/OR signature — would falsify the claim that κ_{ret}/K is the only boundary variable. (ii) A demonstration that outcome commitment occurs in a regime where the phase is provably slaved (e.g. registration without any mode crossing its layer) would falsify the identification of the cut with the lock. (iii) Boundary phenomenology (deviation statistics, lock-time distributions) that fails to collapse onto the single variable κ_{ret}/K across platforms would falsify the crossover structure. (iv) Inherited from Paper 1: a controlled open-system calculation showing the slaved-phase statement fails under field quantization would remove this paper's foundation — we flag it as the live theoretical risk, not a rhetorical one. (v) [v0.4] The location claim has now passed its simplest exact test: a linear absorber with a dense record channel whose recoverability crossover sat far from $\kappa_{\text{ret}}/K = 1$, or showed no crossover, would have falsified it in the smallest model; it sat at 1.3–2.0. A self-sustaining absorber whose crossover moved away from that location would reopen it.

9. Discussion

Relation to the settled literature. This paper is structured as a completion, not a rival, of three bodies of work. Decoherence theory supplies stage (2) entire; we adopt its pointer bases, its timescales, and its own verdict that selection lies outside its scope. Allahverdyan–Balian–Nieuwenhuizen supply registration; their Curie–Weiss apparatus is, in our language, the machinery *below* the layer, and their explicit assumption of a selected asymmetry is the interface our stage (3) fills. Von Neumann's movability theorem is preserved with its domain finally drawn. The genuinely contested relationship is with the programs that also claim stage (3): against engineered-noise reduction models we claim the same phenomenological structure with the noise identified and the fairness proved rather than stipulated (Paper 1), at the price of conditionality on detector premises; against gravitational reduction we claim a mechanism and a non-gravitational, non-universal threshold — the cut is at $\kappa_{\text{ret}} \sim K$, which gravity does not set.

Open problems, in the panel's canonical phrasings. (i) *The single continuous model* (GPT-5.6): the sub-threshold linear equation contains no K; construct one dynamical model, derived from an explicit $H = H_{\text{field}} + H_{\text{absorber}} + H_{\text{bath}} + H_{\text{int}}$, valid on both sides of the boundary, exhibiting which mode becomes marginal at $\kappa_{\text{ret}}/K = 1$ and how the phase degree of freedom and its saturation emerge from absorber microphysics — this subsumes the κ_{ret}-ansatz derivation ([Paper 1] §9.4(viii)) and the lock's closed dynamics in one program. (ii) *The vacuum-restored-phase question* (Grok): does a fully quantized common field restore an effective free phase below the classical threshold? — the decisive inherited risk, now stated as a precise calculation. (iii) The quantitative crossover profile: Figure 1 is a schematic oscillator analogy; the first-principles winding-rate and slip-time profiles for a physical absorber (P4(i) level structure explicit) would convert §8.2's protocol from sketch to prediction, including the framework curve its null model currently lacks. (iv) The K–Γ prefactor for one concrete absorber (§3.2). (v) The track-cascade statistics of §4.2: Mott's conditional stakes composed with per-grain Born games should reproduce track phenomenology; a worked calculation would make the taxonomy's second row as quantitative as its first. (vi) The crossover's completion in the framework paper [Paper 3]. (vii) A dedicated related-work clearance against the synergetics/laser-threshold literature (Haken school): if the qualitative "measurement as threshold crossing" claim has precedent there, the delta — the width formula, the movability-domain theorem, the detector taxonomy — must be stated against it explicitly; flagged by the internal review, not yet performed.

The one-sentence summary. The Heisenberg cut is the locking layer of the detector: free to place wherever linearity reigns — which is a theorem — and pinned, with computable width $w = K/\omega$, exactly where it does not; the classical world begins where the phase stops being slaved, and that address is different, calculable, and testable in every laboratory in physics.

Appendix A — Figure methods

The figure is a **schematic oscillator analogy** (a classical stochastic model with gain and saturation put in by hand), not a simulation of a detector; §3.1 and §9(i) state what a detector-derived version requires. Panel (a): injected Stuart–Landau oscillator $\dot{a} = (g - |a|^2)a - i\Delta a + F + \xi(t)$, $\Delta = 1.0$, $F = 0.35$, complex white noise of strength $D = 10^{-4}$; Euler–Maruyama, $dt = 0.01$; transient $T = 200$ discarded, winding accumulated over $T = 800$; 26 gain values $g \in [-1, 1.5]$. **Two boundaries, both marked:** the Hopf threshold at $g = 0$ (free phase exists) and the injection-locking boundary $g = (F/\Delta)^2 = 0.1225$ (free phase runs; overlay $\sqrt{\Delta^2 - F^2/g}$ valid only there, with limit-cycle amplitude $\sqrt{g}$ and effective locking range $K_{\text{eff}} = F/\sqrt{g}$). At the stated noise the measured running onset is $g \approx 0.35$ — the boundary is noise-smeared upward, and the reviewed draft's caption, which placed "the threshold" at $g = 0$, conflated the two boundaries (corrected in v0.3 after the review panel's independent re-run of this script). *Corrected again in v0.4 (2026-09-04), from a re-run of this model with the noise switched off entirely (`heisenberg_cut_recoverability/stage2_stuart_landau_sharpness.py`, predictions on record): the onset near $g \approx 0.35$ is deterministic. The 50 % winding onset is at $g = 0.355$ with $D = 0$ and 0.352 with $D = 10^{-4}$; noise moves it by 0.003, not from 0.1225 to 0.35. The Adler estimate $g = (F/\Delta)^2$ assumes weak injection, $F \ll \sqrt{g}$, which fails here ($F/\sqrt{g} = 1$ at that point). What actually happens: the locked state is the forced fixed point, $F^2 = u[(g-u)^2 + \Delta^2]$ with $u = |a|^2$; a saddle-node would need $(g-u)(g-3u) = -\Delta^2$, impossible for $g < \sqrt{3}$; the fixed point instead loses stability by a Hopf bifurcation at $u = g/2$, i.e. $g_H = 0.241$ for these parameters, and the winding rate turns on only once the newborn cycle has grown to encircle the origin, near $g \approx 0.35$. The stored energy $|a|^2$ stays at the field-set value $F^2/\Delta^2 = 0.1225$ across both $g = 0$ and g_H (slope 0.004 per unit g) and rises with slope 1.1 only beyond the winding onset: continuous, with a kink, no jump — the energy bookkeeping changes hands from field to site at the running onset, not at the Hopf, and does so smoothly. A counter-rotating drive term ($Fe^{2i\omega_d t}$, $\omega_d = 40$ and 10) changes none of these numbers to three decimals; a record channel $\Gamma_{\text{rec}} = 0.1$ shifts the onset to 0.397 and nothing else. The overlay $\sqrt{\Delta^2 - F^2/g}$ remains the weak-injection asymptote and is valid only well above the onset.* Convergence: below the Hopf threshold the measured winding rate is $\lesssim 4 \times 10^{-5}$ (noise floor at these D, dt, and averaging time; halving dt moves no plotted point by more than the marker size). Panel (b): Adler equation $\dot{\varphi} = \Delta\omega - K \sin\varphi$, $K = 1$; slip period measured as the time for one full 2π advance, $dt = 5 \times 10^{-4}$; 25 detunings $\Delta\omega/K \in [1.02, 5]$; analytic curve $2\pi/\sqrt{\Delta\omega^2 - K^2}$; at $\Delta\omega/K = 5$ simulation and analytic agree to 4×10^{-4} relative. Reproduction: `python3 threshold_clock.py` in `paper2_sims/` (numpy + matplotlib; seed pinned, 20260728), output `figures/p2_fig1_threshold.png`; the repository commit accompanying this revision is the immutable reference.

References

(Author–year style, alphabetical. Entries marked [P1-verified] were bibliographically verified during the companion paper's citation passes; all remaining entries were verified against publisher/indexer records on 2026-07-28 — the verification log, with per-entry source URLs, is `PAPER2_citation_verification_2026-07-28.md` alongside this draft.)

- Adler, R. (1946). A study of locking phenomena in oscillators. *Proc. IRE* **34**, 351–357.
- Adler, S. L. (2003). Why decoherence has not solved the measurement problem: a response to P. W. Anderson. *Stud. Hist. Phil. Mod. Phys.* **34**, 135–142.
- Afzelius, M., Simon, C., de Riedmatten, H., & Gisin, N. (2009). Multimode quantum memory based on atomic frequency combs. *Phys. Rev. A* **79**, 052329. [P1-verified]
- Allahverdyan, A. E., Balian, R., & Nieuwenhuizen, T. M. (2013). Understanding quantum measurement from the solution of dynamical models. *Phys. Rep.* **525**, 1–166. [P1-verified]
- Arndt, M., Nairz, O., Vos-Andreae, J., Keller, C., van der Zouw, G., & Zeilinger, A. (1999). Wave–particle duality of C_{60} molecules. *Nature* **401**, 680–682.
- Bassi, A., & Ghirardi, G. C. (2003). Dynamical reduction models. *Phys. Rep.* **379**, 257–426. [P1-verified]
- Bassi, A., Lochan, K., Satin, S., Singh, T. P., & Ulbricht, H. (2013). Models of wave-function collapse, underlying theories, and experimental tests. *Rev. Mod. Phys.* **85**, 471–527.
- Bell, J. S. (1990). Against "measurement". *Physics World* **3**(8), 33–40.

- Brune, M., Hagley, E., Dreyer, J., Maître, X., Maali, A., Wunderlich, C., Raimond, J. M., & Haroche, S. (1996). Observing the progressive decoherence of the "meter" in a quantum measurement. *Phys. Rev. Lett.* **77**, 4887–4890.
- Camilleri, K., & Schlosshauer, M. (2015). Niels Bohr as philosopher of experiment: does decoherence theory challenge Bohr's doctrine of classical concepts? *Stud. Hist. Phil. Mod. Phys.* **49**, 73–83.
- de Riedmatten, H., Afzelius, M., Staudt, M. U., Simon, C., & Gisin, N. (2008). A solid-state light–matter interface at the single-photon level. *Nature* **456**, 773–777. [P1-verified]
- Dolde, K., Ganapathy, K., Zheng, X., Ma, D., Beloy, K., & Kolkowitz, S. (2025). Direct measurement of the $^3P_0$ clock state natural lifetime in ^{87}Sr. *Phys. Rev. A* **112**, 023121.
- Emary, C., Lambert, N., & Nori, F. (2014). Leggett–Garg inequalities. *Rep. Prog. Phys.* **77**, 016001.
- Equall, R. W., Cone, R. L., & Macfarlane, R. M. (1995). Homogeneous broadening and hyperfine structure of optical transitions in Pr^{3+}:Y_2SiO_5. *Phys. Rev. B* **52**, 3963–3969.
- Fein, Y. Y., Geyer, P., Zwick, P., Kiałka, F., Pedalino, S., Mayor, M., Gerlich, S., & Arndt, M. (2019). Quantum superposition of molecules beyond 25 kDa. *Nat. Phys.* **15**, 1242–1245.
- Ghirardi, G. C., Rimini, A., & Weber, T. (1986). Unified dynamics for microscopic and macroscopic systems. *Phys. Rev. D* **34**, 470–491.
- Hackermüller, L., Hornberger, K., Brezger, B., Zeilinger, A., & Arndt, M. (2004). Decoherence of matter waves by thermal emission of radiation. *Nature* **427**, 711–714.
- Hahn, E. L. (1950). Spin echoes. *Phys. Rev.* **80**, 580–594. [P1-verified]
- Hornberger, K., Gerlich, S., Haslinger, P., Nimmrichter, S., & Arndt, M. (2012). Colloquium: Quantum interference of clusters and molecules. *Rev. Mod. Phys.* **84**, 157–173.
- Joos, E., & Zeh, H. D. (1985). The emergence of classical properties through interaction with the environment. *Z. Phys. B* **59**, 223–243.
- Kjaergaard, M., Schwartz, M. E., Braumüller, J., Krantz, P., Wang, J. I.-J., Gustavsson, S., & Oliver, W. D. (2020). Superconducting qubits: current state of play. *Annu. Rev. Condens. Matter Phys.* **11**, 369–395.
- Könz, F., Sun, Y., Thiel, C. W., Cone, R. L., Equall, R. W., Hutcheson, R. L., & Macfarlane, R. M. (2003). Temperature and concentration dependence of optical dephasing, spectral-hole lifetime, and anisotropic absorption in Eu^{3+}:Y_2SiO_5. *Phys. Rev. B* **68**, 085109.
- Kuhlmann, A. V., Prechtel, J. H., Houel, J., Ludwig, A., Reuter, D., Wieck, A. D., & Warburton, R. J. (2015). Transform-limited single photons from a single quantum dot. *Nat. Commun.* **6**, 8204.
- Kurnit, N. A., Abella, I. D., & Hartmann, S. R. (1964). Observation of a photon echo. *Phys. Rev. Lett.* **13**, 567–568. [P1-verified]
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM J. Res. Dev.* **5**, 183–191.
- Leggett, A. J., & Garg, A. (1985). Quantum mechanics versus macroscopic realism: is the flux there when nobody looks? *Phys. Rev. Lett.* **54**, 857–860.
- Levstein, P. R., Usaj, G., & Pastawski, H. M. (1998). Attenuation of polarization echoes in nuclear magnetic resonance: a study of the emergence of dynamical irreversibility in many-body quantum systems. *J. Chem. Phys.* **108**, 2718–2724.
- Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E., & Schmidt, P. O. (2015). Optical atomic clocks. *Rev. Mod. Phys.* **87**, 637.
- Mott, N. F. (1929). The wave mechanics of α-ray tracks. *Proc. R. Soc. A* **126**, 79–84.
- Ollivier, H., Poulin, D., & Zurek, W. H. (2004). Objective properties from subjective quantum states: environment as a witness. *Phys. Rev. Lett.* **93**, 220401.
- [Paper 1] Claude Fable 5 (Anthropic), & Bramble, J. M. (2026). A field–matter selector for outcome production, conditioned on a detector ready-state measure. Manuscript v0.7, with published AI peer-review record. [Framework repository] *(Entry bumped from v0.5.2 on 2026-08-01, tri-paper consistency ledger item 23. Retitled and bumped to v0.7 on 2026-08-31, tri-paper round 2 item L2-4: the earlier title, "The Born rule as a derived fair game," was withdrawn in that revision along with the derivation claim it asserted.)*
- Pastawski, H. M., Levstein, P. R., Usaj, G., Raya, J., & Hirschinger, J. (2000). A nuclear magnetic resonance answer to the Boltzmann–Loschmidt controversy? *Physica A* **283**, 166–170.
- Penrose, R. (1996). On gravity's role in quantum state reduction. *Gen. Rel. Grav.* **28**, 581–600.

- Pikovsky, A., Rosenblum, M., & Kurths, J. (2001). *Synchronization: A Universal Concept in Nonlinear Sciences.* Cambridge University Press.
- Sabbah, A. J., & Riffe, D. M. (2002). Femtosecond pump–probe reflectivity study of silicon carrier dynamics. *Phys. Rev. B* **66**, 165217.
- Schlosshauer, M. (2004). Decoherence, the measurement problem, and interpretations of quantum mechanics. *Rev. Mod. Phys.* **76**, 1267–1305.
- Schlosshauer, M., & Camilleri, K. (2008). The quantum-to-classical transition: Bohr's doctrine of classical concepts, emergent classicality, and decoherence. arXiv:0804.1609.
- Steck, D. A. (2025). Rubidium 87 D Line Data, revision 2.3.4 (8 August 2025). https://steck.us/alkalidata
- Tegmark, M. (1993). Apparent wave function collapse caused by scattering. *Found. Phys. Lett.* **6**, 571–590.
- von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik.* Springer. English translation: *Mathematical Foundations of Quantum Mechanics*, Princeton University Press (1955).
- Zeh, H. D. (1970). On the interpretation of measurement in quantum theory. *Found. Phys.* **1**, 69–76.
- Zurek, W. H. (1981). Pointer basis of quantum apparatus: into what mixture does the wave packet collapse? *Phys. Rev. D* **24**, 1516–1525.
- Zurek, W. H. (1982). Environment-induced superselection rules. *Phys. Rev. D* **26**, 1862–1880.
- Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.* **75**, 715–775.
- Zurek, W. H. (2009). Quantum Darwinism. *Nature Phys.* **5**, 181–188.
- [Framework repository] Bramble, J. M., & Claude (Anthropic). *DiracKuramotoFramework.* Public version-controlled repository. https://github.com/rayolddog/DiracKuramotoFramework (citation form per the repository's CITATION.cff and AUTHORSHIP.md authorship policy v2.0, harmonized 2026-07-28)
- [Paper 3, in preparation] The Dirac–Kuramoto framework paper (revision of the Two-Regimes manuscript), which develops the dissipative sector's foliation and its experimental program.